WORKED SOLUTION

STEP II 2005, Question 9

Mechanics • Friction

STEP II 2005, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q9

The key idea

Treat the two particles separately. A has no applied force, so its equation fixes the tension the string must provide. All the freedom is in the *direction* of P on B. Tilting P changes the normal reaction on B, and so the friction on B. Tilting P off the slope wastes some of its component along the slope but changes the friction, and the best tilt balances these two effects. It comes out as an expression $a\cos\theta + b\sin\theta$ to maximise.

Solution

We model A and B as particles and the string as light and inextensible, lying along ℓ. We take P to act in the vertical plane containing ℓ. A component of P across the slope would do nothing to help. Note that

$$\sin 30^\circ = \tfrac{1}{2}, \qquad \cos 30^\circ = \tfrac{\sqrt{3}}{2}, \qquad \tan 30^\circ = \frac{1}{\sqrt{3}} = \tfrac{1}{3}\sqrt{3}.$$

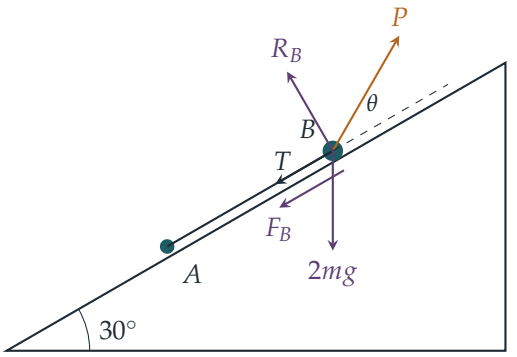
So $\mu_B = \tan 30^\circ$: on its own, B would be exactly on the point of slipping. Also $\mu_A = \frac{1}{2}\tan 30^\circ$, so A on its own would slip.

Particle A. Resolving perpendicular to the plane, $R_A = mg\cos 30^\circ = \frac{\sqrt{3}}{2}mg$, so the greatest friction on A is

$$\mu_A R_A = \tfrac{\sqrt{3}}{6}\cdot\tfrac{\sqrt{3}}{2}mg = \tfrac{1}{4}mg.$$

The weight component down the slope is $mg\sin 30^\circ = \frac{1}{2}mg$.

Particle B. Let P make an angle θ with the upward direction of ℓ, measured away from the plane. (A negative θ means P pushes into the plane.) Let T be the tension.



The diagram shows friction F_B acting down the slope, as in part (i). Perpendicular to the plane,

$$R_B + P\sin\theta = 2mg\cos 30^\circ \quad\Longrightarrow\quad R_B = \sqrt{3}\,mg - P\sin\theta.$$

So the greatest friction available at B is

$$\mu_B R_B = \tfrac{1}{\sqrt{3}}\left(\sqrt{3}\,mg - P\sin\theta\right) = mg - \tfrac{1}{\sqrt{3}}P\sin\theta.$$

The weight component of B down the slope is $2mg\sin 30^\circ = mg$.

Part (i)

For the particles to move up ℓ, friction on each acts down the slope. At the least P both are on the point of moving, so friction is limiting on both.

For A to move up, the string must overcome the weight component and the friction:

$$T = \tfrac{1}{2}mg + \tfrac{1}{4}mg = \tfrac{3}{4}mg.$$

For B, along the slope, the component of P must overcome T, the weight component and limiting friction:

$$P\cos\theta = \tfrac{3}{4}mg + mg + \left(mg - \tfrac{1}{\sqrt{3}}P\sin\theta\right),$$

$$P\left(\cos\theta + \tfrac{1}{\sqrt{3}}\sin\theta\right) = \tfrac{11}{4}mg.$$

Ask yourself Why does tilting P away from the slope help, when it reduces the component of P along the slope?

Lifting B slightly reduces R_B, and so the friction that P has to beat. For a small tilt θ, $\cos\theta$ falls only like $1 - \frac{1}{2}\theta^2$ while $\sin\theta$ grows like θ. So at first the gain from reduced friction wins. To find the best angle, write

$$\cos\theta + \tfrac{1}{\sqrt{3}}\sin\theta = \tfrac{2}{\sqrt{3}}\left(\tfrac{\sqrt{3}}{2}\cos\theta + \tfrac{1}{2}\sin\theta\right) = \tfrac{2}{\sqrt{3}}\cos(\theta - 30^\circ).$$

This is at most $\frac{2}{\sqrt{3}}$, with equality exactly when $\theta = 30°$. Since P times this factor must equal $\frac{11}{4}mg$, P is least when the factor is greatest:

$$P_{\min} = \tfrac{11}{4}mg \cdot \tfrac{\sqrt{3}}{2} = \tfrac{11}{8}\sqrt{3}\,mg,$$

as required. We must check that B stays on the plane:

$$R_B = \sqrt{3}\,mg - \tfrac{11}{8}\sqrt{3}\,mg \cdot \tfrac{1}{2} = \tfrac{5}{16}\sqrt{3}\,mg > 0.$$

So P acts in the vertical plane through ℓ, at 30° to ℓ on the side away from the plane. That is, it acts **at** 60° **above the horizontal**, pulling up the slope. (Notice that the best angle with the slope is $\arctan \mu_B$, the angle of friction for B.)

Part (ii)

Now P only has to stop the system slipping down, so friction on each particle acts up the slope. Without P, A cannot stay at rest alone, because its weight component $\frac{1}{2}mg$ exceeds the greatest friction $\frac{1}{4}mg$. So the string must hold A back:

$$T \geq \tfrac{1}{2}mg - \tfrac{1}{4}mg = \tfrac{1}{4}mg.$$

In particular the string must be taut. For B not to slip down, the forces up the slope must be able to balance T and the weight component:

$$P\cos\theta + \left(mg - \tfrac{1}{\sqrt{3}}P\sin\theta\right) \geq T + mg.$$

The left-hand side does not depend on T, so the least P comes with the least tension, $T = \frac{1}{4}mg$. Then

$$P\left(\cos\theta - \tfrac{1}{\sqrt{3}}\sin\theta\right) \geq \tfrac{1}{4}mg.$$

Now $\cos\theta - \frac{1}{\sqrt{3}}\sin\theta = \frac{2}{\sqrt{3}}\cos(\theta + 30°) \leq \frac{2}{\sqrt{3}}$, with equality when $\theta = -30°$. This time the best direction pushes B *into* the plane, which increases R_B and so the friction helping to hold B. Hence the least magnitude is

$$P_{\min} = \tfrac{1}{4}mg \cdot \tfrac{\sqrt{3}}{2} = \tfrac{1}{8}\sqrt{3}\,mg.$$

It acts at 30° below the upward direction of ℓ. Since ℓ rises at 30°, that direction is horizontal: a horizontal push on B towards the slope. At this value, A and B are both in limiting equilibrium, with $T = \frac{1}{4}mg$.

Checks

- We checked numerically with $m = g = 1$. A scan over every direction of P in the vertical plane found the least P for each one by bisection. In (i) the minimum is $2.3816 \approx \frac{11}{8}\sqrt{3}$, at 30.0° to the slope. In (ii) it is $0.2165 \approx \frac{1}{8}\sqrt{3}$, at −30.0°, which is horizontal. We also confirmed that a full equilibrium exists at the minimum in (ii).

- Total bookkeeping for (i) with P along the slope ($\theta = 0$): we need $P = \frac{1}{2}mg + \frac{1}{4}mg + mg + mg = \frac{11}{4}mg \approx 2.75mg$. This is larger than $2.38mg$, as it should be.
- For (ii) with P along the slope, the system needs $\frac{3}{2}mg - \frac{1}{4}mg - mg = \frac{1}{4}mg$. The horizontal push does better, at $\frac{\sqrt{3}}{8}mg \approx 0.217mg$.

Common slips

- Assuming P acts along the slope. The question asks for the least *magnitude*, so the direction is yours to choose.
- Forgetting that $P\sin\theta$ changes R_B, and hence the limiting friction on B.
- In (ii), keeping the friction directions from (i). When the danger is slipping down, friction acts up the slope.

Beyond STEP

Both parts rest on the same principle: the most efficient force against friction makes the angle of friction $\lambda = \arctan\mu$ with the line of motion. In (i) it is tilted away from the plane, and in (ii) it is tilted into it. This is because the normal reaction and limiting friction combine into a single "total reaction" at angle λ to the normal. Thinking in terms of this total reaction turns many friction problems into triangle-of-forces geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and constrained motion carefully at first-year level.

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